

VSB ENGINEERING COLLEGE

Academic Performance & Parent Notification Platform

ADMINISTRATOR OPERATING MANUAL — INSTITUTIONAL COMMAND CENTER

1. Administrator Access & Authentication

The Administrator Command Center provides institutional oversight across all departments, faculty rosters, student batches, mark modification audit trails, and college-wide parent notification dispatching.

1 Access the Live Portal URL

Open your web browser (Chrome, Edge, Firefox, or Mobile) and navigate to:
Live URL: <https://remarkable-exploration-production-ab2a.up.railway.app> (or local <http://localhost:5173>)

2 Select the Admin Login Tab

On the login screen, click the Deep Indigo tab labeled "Admin Login". (Note: The system blocks staff from logging in via the Admin tab and vice-versa for strict role isolation).

3 Sign In with Master Administrator Credentials

Email: admin@college.edu | Password: admin123
Click "Sign In as Administrator". The system validates credentials and redirects you to `/admin/dashboard`.

2. Navigating the Admin Command Center

The Admin Dashboard displays key real-time metrics summarizing the active academic status of VSB Engineering College:

- **Key Performance Cards:** Displays Total Enrolled Students, Total Teaching Faculty, Active Departments, Total Notifications Sent, and College Pass Percentage.
- **Department Performance Charts:** Visual bar graphs comparing pass rates across Computer Science, Electronics, Mechanical, and Civil Engineering.
- **Quick Action Shortcuts:** Direct 1-click links to "Bulk Import Students", "Add Faculty", "Create Notice", and "Audit Logs".

3. Setting Up the Academic Structure (Hierarchy)

Before enrolling students or assigning faculty, the institutional structure must be configured once per academic year:

- **Departments Management:** In the sidebar, click "Academic Structure" -> "+ Add Department". Enter Department Code (e.g. "CSE") and Title (e.g. "Computer Science & Engineering").
- **Academic Year Session:** Set the active year (e.g. "2025-2026"). Marking a session active automatically associates new sections with this cohort.
- **Undergraduate Years:** Verify the 4 standard study years: Year 1 (First Year), Year 2 (Second Year), Year 3 (Third Year), Year 4 (Final Year).
- **Section Allocations:** Add Sections (e.g., Section A, Section B) under each Department and Year.
- **Curriculum Subjects:** Click "+ Add Subject". Input Subject Code (e.g., CS8501), Subject Name (e.g., "Theory of Computation"), Semester (5), and Maximum Marks (default: 100.0).

4. Faculty Management & Class Assignment (Teacher Mapping)

The Administrator controls faculty onboarding, department assignments, and subject handling rights:

Step 4.1: Adding a New Faculty Member

1 Open Faculty Directory

In the left navigation sidebar, click "Faculty Management" -> Click the blue "+ Add Faculty" button.

2 Input Faculty Profile Details

- Full Name: e.g., Prof. Sarah Jenkins
- Official Email: sarah.cse@college.edu | Username: sarah.jenkins
- Department: Select Computer Science & Engineering | Date of Birth: 1988-08-15

3 Standardized Default Password Generation

The system automatically provisions a standardized password using the formula [FirstName]@[DDMMYYYY] (e.g., Sarah@15081988). Inform the faculty member of their initial credentials.

Step 4.2: Assigning Faculty to Classes (Teacher Assignment)

Teaching staff can ONLY view and grade classes explicitly assigned to them by an Administrator:

- **Initiate Assignment:** On the Faculty list, click "Assign Subject" next to the instructor.
- **Select Class Cohort:** Choose: Department (CSE) -> Year (Third Year) -> Section (A) -> Subject (CS8501).
- **Confirm Mapping:** Click "Confirm Assignment". The instructor can now immediately log in, see Section 3-A, and enter marks.

5. Student Management & Bulk Upload (Excel / CSV)

Administrators can register students individually or upload entire class cohorts in bulk within seconds:

Method A: Bulk Cohort Import via Excel / CSV (Recommended)

- **Step 1: Download Sample Template:** Navigate to "Student Directory" -> Click "Bulk Import" -> Click "Download Sample CSV Template".
- **Step 2: Prepare Roster Spreadsheet:** Ensure the spreadsheet has these mandatory columns:

Register Number	Student Name	Dept	Year	Sec	Parent Name	Parent Mobile
921321104001	Aravind S	CSE	3	A	Senthil Kumar	9876543210
921321104002	Divya M	CSE	3	A	Murugan P	9876543211
921321104003	Karthik R	CSE	3	A	Radhakrishnan K	9876543212

- **Step 3: Select Duplicate Resolution Mode:** • "Skip Existing": Leaves already-enrolled students untouched, importing only new registers.
- "Update Existing": Updates existing student contact records with new phone numbers or section transfers.
- **Step 4: Execute & Verify:** Drag & drop the file into the drop zone. The system checks telephone and register numbers and displays an instant validation summary.

Method B: Add Single Student

Click "Student Directory" -> "+ Add Student". Fill in Register Number, Full Name, Department, Year, Section, Parent Name, and 10-digit Parent Mobile -> Click "Save Student".

6. Mark Oversight, Modification & Audit Trail (NBA / NAAC Compliance)

Administrators maintain institutional integrity by monitoring student marks, pass distributions, and grade corrections:

Step 6.1: Reviewing Class Mark Sheets

- **Access Class Sheets:** In sidebar, click "Marks & Grades". Select Department, Year, Section, Subject, and Assessment.
- **Performance Breakdown:** View marks obtained, percentage score, letter grade, and PASS / CRITICAL status badges.

Step 6.2: Grade Corrections & Mandatory Justification

If an evaluation error occurred and a score needs alteration after initial submission:

- **Edit Score:** Click "Edit" on the student row and modify the mark.
- **Mandatory Justification Prompt:** The system requires entering a "Reason for Modification" (e.g. "Re-evaluation re-totalling correction approved by Dean").
- **Immutable Audit Logging:** The previous score, revised score, modifying staff/admin identity, exact timestamp, and justification are permanently stored in the audit database.

Step 6.3: Inspecting the Audit Trail Viewer

- **Audit Trail Menu:** Click "Audit Logs" in the sidebar to review a tamper-proof chronological ledger of all administrative logins, mark entries, student imports, and circular dispatches.

7. Broadcasting College Notices & Circulars (SMS & WhatsApp)

Publish announcements, exam schedules, and holiday circulars directly to parents and students:

1 Click "+ Create Notice"

In the sidebar, navigate to "College Notices" -> Click the "+ Create Notice" button.

2 Select Notice Classification

Choose category: "Exam Timetable", "Holiday Circular", "College Reopening", "Academic Fee", or "Urgent Announcement".

3 Set Audience Scope & Channels

- Target Scope: Choose "All College", "Specific Department", or "Specific Section".
- Delivery Channel: Select "SMS", "WhatsApp", or "Both".

4 Publish & Dispatch

Type message content -> Click "Publish & Send Notifications". The queue automatically dispatches alerts to parent mobile phones.

8. Institutional Reports & PowerPoint Presentation Export

Generate formal documentation for management reviews, accreditation audits, and Board of Studies meetings:

- **Official PDF Grade Sheets:** Go to "Reports & Analytics" -> Select Department, Year & Exam -> Click "Download PDF". Produces signed institutional grade rosters.
- **Executive PowerPoint Export (.pptx):** Click "Export to PowerPoint (.pptx)". Generates automated meeting slides complete with department pass percentages, subject toppers, and critical lists.

9. Administrator Semester Launch Checklist (Week 1 to Exam Week)

Follow this chronological SOP checklist to manage academic operations smoothly throughout the semester:

Phase 1: Pre-Semester Setup (Weeks 1 - 2)

- [] **Academic Year & Structure:** Ensure the current Academic Year (2025-2026) is active. Verify all Departments and Sections.
- [] **Curriculum & Subjects:** Verify syllabus subject codes, semester mappings, and maximum mark allocations.
- [] **Faculty Onboarding:** Add new teaching faculty and verify standardized login passwords (FirstName@DDMMYYYY).
- [] **Teacher Assignments:** Map every subject and section to its designated subject handler.
- [] **Student Cohort Import:** Upload new student enrollments using the Bulk Excel Import tool.

Phase 2: Mid-Semester Assessments (Weeks 6 - 12)

- [] **Assessment Scheduling:** Verify Internal Assessment 1 (IA-1), IA-2, and Model Exam entries.
- [] **Faculty Grading Deadlines:** Monitor faculty mark entry progress across departments.
- [] **Audit Mark Alterations:** Review mark changes and reasons in the Audit Trail viewer.
- [] **Parent Notifications:** Trigger automated SMS/WhatsApp alerts to notify parents of test performance.

Phase 3: Semester Review & Accreditation (Weeks 14 - 16)

- [] **Department Analytics Review:** Inspect cross-department pass/fail distributions.
- [] **Executive Reporting:** Export PowerPoint (.pptx) decks for HOD and Principal review.
- [] **Archival:** Export finalized consolidated PDF grade sheets for examination cell records.

10. Administrator Troubleshooting & Support

- **Forgotten Faculty Password:** Administrators can reset any faculty password by editing the faculty member profile under "Faculty Management".
- **Student Phone Number Change:** Locate student in "Student Directory", click "Edit", update parent mobile number, and click "Save".
- **Cloud Database Backup:** The PostgreSQL database runs automatically on Railway with automated persistence.

Administrator Security Notice

Always keep administrator credentials secure. Never share master passwords. If logging in from a public lab terminal, always click "Log Out" at the bottom of the sidebar when finished.